

Harshil Singh

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Education

Indian Institute of Technology, Bhubaneswar
B.Tech Computer Science and Engineering
Bal Bharati School
Senior School Certificate Examination, CBSE
JEE Advanced

GPA: 7.49
Nov. 2022 - present
Percentage: 91.4%
April 2019 - March 2021
Rank: 2487

Projects

- BiasFree - Tredence Hackathon Project**  *Machine Learning*
- Model Creation:** Constructed a ML model to detect gender-biased sentences in paragraphs.
 - Model Architecture:** Designed and executed a sophisticated model architecture utilizing Random Forest as the meta-learner, which enhanced overall model performance and reduced misclassification rates by 20% in real-time user interactions.
 - Performance:** Achieved a 90% AUC-ROC score, demonstrating high-accuracy bias detection across varied text data.
- CricPreditt - Cricket Score application with Live win Prediction**  *Machine Learning*
- Model Creation:** Created a deep learning predictor for cricket win probabilities using 100,000+ rows of data.
 - Performance:** Trained using neural networks with TensorFlow/Keras and Scikit-learn, reaching 87.5% accuracy.
 - Data Processing:** Applied Pandas and Matplotlib to analyze data and visualize trends, cutting latency by 20%.

Course Projects

- Cloud particles image clustering**  *Unsupervised Learning*
- Data preparation:** Isolated individual cloud particles and preprocessed to create a high-quality dataset of around 18k images.
 - Modeling:** Adopted ViT-Base Patch/224 for feature extraction, followed by UMAP and K-Means clustering to identify 7 distinct groups.
 - Performance Evaluation:** Accomplished strong clustering performance with Silhouette Score 43%, Davies-Bouldin Index 0.81, and Calinski-Harabasz Score 29,874.
- RISC-V CPU Simulator - Assembly to Machine Code Converter** *Computer Architecture*
- Model Development:** Designed a model to convert assembly language code into machine code and simulate a RISC-V architecture CPU.
 - Programming:** Implemented in C/C++ to process and execute RISC-V instructions with high fidelity.
 - Simulation:** Achieved accurate CPU simulation, reducing execution errors by 30% through optimized instruction handling.

Certification and Course Projects

- Supervised Machine Learning: Regression and Classification** *Certificate*
DeepLearning.AI, Stanford University
May 2024
- Mastered regression/classification (90%+ accuracy) and preprocessed 10,000+ entries, managing missing values and scaling.
- Advanced Learning Algorithms** *Certificate*
DeepLearning.AI, Stanford University
May 2024
- Built TensorFlow neural networks for multi-class classification (92% accuracy) and mastered decision trees/ensembles (random forests, boosted trees), improving performance by 25% over baseline.
- Unsupervised Learning, Recommenders, Reinforcement Learning** *Certificate*
DeepLearning.AI, Stanford University
May 2024
- Learned unsupervised techniques (clustering, anomaly detection), applied them to 50,000+ data points (95% precision in outlier detection), and trained a reinforcement learning model, boosting rewards by 40% over 100 iterations.

Key Skills

- Programming Languages:** C/C++, Python, MySQL
- Tools:** MatLab, Jupyter Notebook
- Framework/Libraries:** TensorFlow, Keras, Numpy, MatPlotLib, Scikit-Learn, Tkinter, Pandas

Position of Responsibility

- Sketching Mentor** (April 2023 - present)
Kalakriti Society, IIT Bhubaneswar
- Mentored 30+ students in sketching techniques, improving their artistic proficiency by 50% (self-reported feedback).
 - Assisted in planning and coordinating society events, exhibitions, and collaborative projects.

Extra-Curricular Activities

- Kalakriti Society-fine arts society, IITBBS (2022 - present)
- RISC-robotics society, IITBBS (2022 - 2023)
- Events Team, Alma Fiesta (Socio-Cultural Fest), IIT BBS. (2022 - 2023)